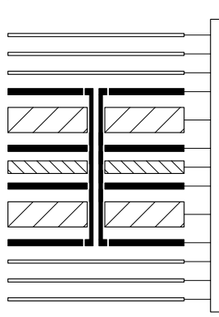
Watch-DevKit-HW Fabrication Document

Layer Stack Legend



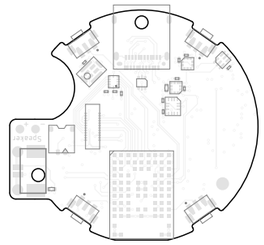
Material	Layer	Thickness	Dielectric	Type	Gerber
	F.Paste			Paste Mask	
	F.Silkscreen			Legend	GBR
	F.Mask	0.01mm		Solder Mask	GBR
Copper	**F.Cu**	**0.035mm (1.00oz)**		**Signal**	**GBR**
Core		0.196mm	FR4	Dielectric	
Copper	**Inner1**	**0.035mm (1.00oz)**		**Plane**	**GBR**
Prepreg		0.43mm	FR4	Dielectric	
Copper	**Inner2**	**0.035mm (1.00oz)**		**Plane**	**GBR**
Core		0.196mm	FR4	Dielectric	
Copper	**B.Cu**	**0.035mm (1.00oz)**		**Signal**	**GBR**
	B.Mask	0.01mm		Solder Mask	GBR
	B.Silkscreen			Legend	GBR
	B.Paste			Paste Mask	

Total thickness: 0.982mm
Note: external layer thicknesses are specified after plating

Impedance Table

Transmission Line	Impedance [ohms]	Tolerance [%]	Layer	Trace Width [mm]	Gap [mm]	Ref. Layers

Top Fabrication (Scale 1:1)



FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

1) FABRICATE PER IPC-6012A CLASS 2.

2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

   DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING FOR REFERENCE ONLY.

3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

   SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING FOR REFERENCE ONLY.

4) SURFACE FINISH: HAL LEAD-FREE

5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR WHITE.

6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING BLACK NON-CONDUCTIVE EPOXY INK.

7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.

8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING

9) PCB MATERIAL REQUIREMENTS:

   A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
   B. Tg 150 C OR EQUIVALENT.

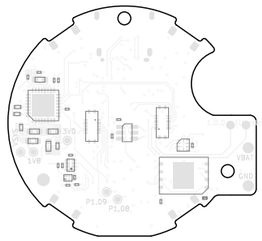
10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

BOARD SIZE	40.477 × 37.500 mm
BOARD THICKNESS	0.982 mm
TRACE WIDTH	0.150 mm
TRACE TO TRACE	0.125 mm
MIN. HOLE (PTH)	0.250 mm
MIN. HOLE (NPTH)	0.650 mm
ANNULAR RING	0.100 mm
COPPER TO HOLE	0.125 mm
COPPER TO EDGE	0.125 mm
HOLE TO HOLE	0.250 mm

Comments: ZSWatch	Company: **ZSWatch**		Variant: PRELIMINARY	Git Hash: e50a45a
	Board Name: **Watch-DevKit-HW**		Project Name: **ZSWatch Watch-DevKit**	
Sheet Title: Top Fabrication (Scale 1:1)	File Name: ZSWatch-Watch.kicad_pcb	Designer: Daniel Kampert	Date: 2025-07-13	Revision: 1.0.0+ (Unreleased)
Sheet Path:		Reviewer:	Size: **A3**	Sheet: **1** of **10**

Watch-DevKit-HW Fabrication Document

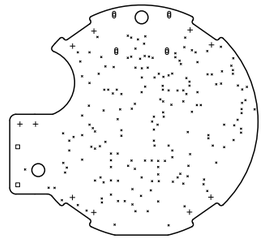
Bottom Fabrication (Scale 1:1)



Comments: ZSWatch	Company: **ZSWatch**		Variant: PRELIMINARY	Git Hash: e50a45a
	Board Name: **Watch-DevKit-HW**		Project Name: **ZSWatch Watch-DevKit**	
Sheet Title: Bottom Fabrication (Scale 1:1)	File Name: ZSWatch-Watch.kicad_pcb	Designer: Daniel Kampert	Date: 2025-07-13	Revision: 1.0.0+ (Unreleased)
	Sheet Path:	Reviewer:	Size: **A3**	Sheet: **2** of **10**

Watch-DevKit-HW Fabrication Document

Drill Drawing L1 - L4 (Scale 1:1)



Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	183	0,25mm (9,84mils)	PTH	Round	F.Cu - B.Cu	Via
○	4	0,50mm (19,69mils)	PTH	Slot	F.Cu - B.Cu	Pad
+	10	0,80mm (31,50mils)	PTH	Round	F.Cu - B.Cu	Pad
□	2	0,90mm (35,43mils)	PTH	Round	F.Cu - B.Cu	Pad
	Total 199					

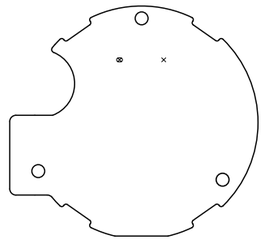
Comments: ZSWatch	Company: **ZSWatch**		Variant: PRELIMINARY	Git Hash: e50a45a	
	Board Name: **Watch-DevKit-HW**		Project Name: **ZSWatch Watch-DevKit**		
Sheet Title: Drill Drawing (L1 - L4)	File Name: ZSWatch-Watch.kicad_pcb	Designer: Daniel Kampert	Date: 2025-07-13	Revision: 1.0.0+ (Unreleased)	
Sheet Path:		Reviewer:	Size: **A3**	Sheet: **3** of **10**	

Watch-DevKit-HW Fabrication Document

Drill Drawing L1 - L4 (Scale 1:1)

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	1 + 1	0,65mm (25,59mils)	NPTH	Round + Slot	F.Cu - B.Cu	Mechanical
○	1	2,10mm (82,68mils)	NPTH	Round	F.Cu - B.Cu	Mechanical
	Total 3					

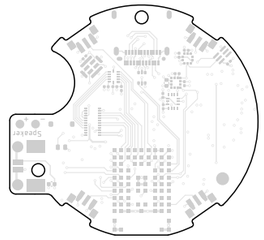


Comments: ZSWatch	Company: **ZSWatch**			Variant: PRELIMINARY	Git Hash: e50a45a
	Board Name: **Watch-DevKit-HW**			Project Name: **ZSWatch Watch-DevKit**	
Sheet Title: Drill Drawing (L1 - L4)	File Name: ZSWatch-Watch.kicad_pcb	Designer: Daniel Kampert		Date: 2025-07-13	Revision: 1.0.0+ (Unreleased)
	Sheet Path:	Reviewer:		Size: **A3**	Sheet: **4** of **10**

Watch-DevKit-HW Fabrication Document

Top Test Points (Scale 1:1)

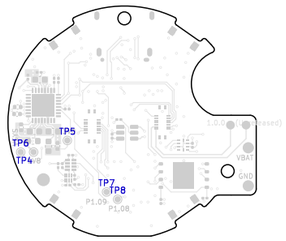
Ref.	Net	X [mm]	Y [mm]	Ref.	Net	X [mm]	Y [mm]
TP1	VBAT	186.80	-142.60				
TP2	Net-(X702-Pin_1),Net-(X702-Pin_3)	189.70	-138.90				
TP3	GND	186.80	-148.80				



Comments: ZSWatch	Company: **ZSWatch**		Variant: PRELIMINARY	Git Hash: e50a45a
	Board Name: **Watch-DevKit-HW**		Project Name: **ZSWatch Watch-DevKit**	
Sheet Title: Top Test Points (Scale 1:1)	File Name: ZSWatch-Watch.kicad_pcb	Designer: Daniel Kampert	Date: 2025-07-13	Revision: 1.0.0+ (Unreleased)
Sheet Path:		Reviewer:	Size: **A3**	Sheet: **5** of **10**

Watch-DevKit-HW Fabrication Document

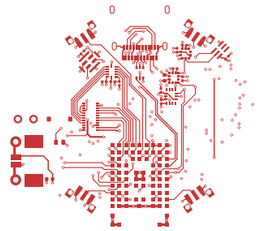
Bottom Test Points (Scale 1:1)



Comments: ZSWatch	Company: **ZSWatch**		Variant: PRELIMINARY	Git Hash: e50a45a
	Board Name: **Watch-DevKit-HW**		Project Name: **ZSWatch Watch-DevKit**	
Sheet Title: Bottom Test Points (Scale 1:1)	File Name: ZSWatch-Watch.kicad_pcb	Designer: Daniel Kampert	Date: 2025-07-13	Revision: 1.0.0+ (Unreleased)
Sheet Path:		Reviewer:	Size: **A3**	Sheet: **6** of **10**

Watch-DevKit-HW Fabrication Document

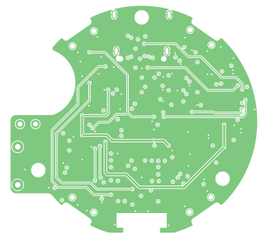
F.Cu (Scale 1:1)



Comments: ZSWatch	Company: **ZSWatch**		Variant: PRELIMINARY	Git Hash: e50a45a
	Board Name: **Watch-DevKit-HW**		Project Name: **ZSWatch Watch-DevKit**	
Sheet Title: F.Cu (Scale 1:1)	File Name: ZSWatch-Watch.kicad_pcb	Designer: Daniel Kampert	Date: 2025-07-13	Revision: 1.0.0+ (Unreleased)
Sheet Path:		Reviewer:	Size: **A3**	Sheet: **7** of **10**

Watch-DevKit-HW Fabrication Document

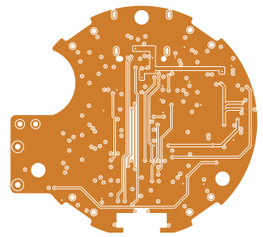
Inner1 (Scale 1:1)



Comments: ZSWatch	Company: **ZSWatch**			Variant: PRELIMINARY	Git Hash: c50a45a
	Board Name: **Watch-DevKit-HW**			Project Name: **ZSWatch Watch-DevKit**	
Sheet Title: Inner1 (Scale 1:1)	File Name: ZSWatch-Watch.kicad_pcb	Designer: Daniel Kampert		Date: 2025-07-13	Revision: 1.0.0+ (Unreleased)
Sheet Path:		Reviewer:		Size: **A3**	Sheet: **8** of **10**

Watch-DevKit-HW Fabrication Document

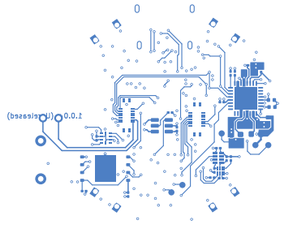
Inner2 (Scale 1:1)



Comments: ZSWatch	Company: **ZSWatch**		Variant: PRELIMINARY	Git Hash: c50a45a
	Board Name: **Watch-DevKit-HW**		Project Name: **ZSWatch Watch-DevKit**	
Sheet Title: Inner2 (Scale 1:1)	File Name: ZSWatch-Watch.kicad_pcb	Designer: Daniel Kampert	Date: 2025-07-13	Revision: 1.0.0+ (Unreleased)
Sheet Path:		Reviewer:	Size: **A3**	Sheet: **9** of **10**

Watch-DevKit-HW Fabrication Document

B.Cu (Scale 1:1)



Comments: ZSWatch	Company: **ZSWatch**		Variant: PRELIMINARY	Git Hash: e50a45a
	Board Name: **Watch-DevKit-HW**		Project Name: **ZSWatch Watch-DevKit**	
Sheet Title: B.Cu (Scale 1:1)	File Name: ZSWatch-Watch.kicad_pcb	Designer: Daniel Kampert	Date: 2025-07-13	Revision: 1.0.0+ (Unreleased)
Sheet Path:		Reviewer:	Size: **A3**	Sheet: **10** of **10**